

Every integer drops into one bucket

Bucket 0
remainder 0 mod 3

..., 3, 6,
9, 12, ...

Bucket 1
remainder 1 mod 3

..., 1, 4,
7, 10, ...

Bucket 2
remainder 2 mod 3

..., 2, 5,
8, 11, ...

Triples whose labels sum to a multiple of 3



$$0 + 0 + 0 \\ = 0$$



$$1 + 1 + 1 \\ = 3$$



$$2 + 2 + 2 \\ = 6$$



$$0 + 1 + 2 \\ = 3$$